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Subject Code: BP604T

Roll No:

BPHARM

(SEM. VI) THEORY EXAMINATION 2025-26

BIOPHARMACEUTICS AND PHARMACOKINETICS – THEORY

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

10x2 = 20

- (a) Mention the factors affecting protein-drug binding.
- (b) Define Nonlinear Pharmacokinetics.
- (c) Mention the non-per-oral extra-vascular routes.
- (d) Define volume of distribution. What is its significance?
- (e) How will you define bioavailability?
- (f) Enumerate the factors affecting drug absorption.
- (g) Classify different types of pharmacokinetic models.
- (h) Define clearance. What are its units?
- (i) Mention the objectives of bioavailability.
- (j) Enlist the various methods to enhance the dissolution rates of poorly soluble drugs.

SECTION B

2. Attempt any two of the following:

2 x 10 = 20

- (a) Describe the active transport mechanisms of drug absorption through GIT.
- (b) What is steady state concentration? How is steady state achieved? Mention the influence of R on steady state.
- (c) Suggest the Clinical significance of protein binding of drugs.

SECTION C

3. Attempt any five part of the following:

7 x 5 = 35

- (a) Elaborate the various factors causing non-linearity.
- (b) Write a detailed note on the calculation of absorption rate constant for extravascular administration.
- (c) Explain in detail the plasma level-time curve for a two compartment open model.
- (d) What do you mean by drug metabolism? Explain factors affecting renal excretion of drugs.
- (e) Discuss the various methods for the assessment of bioavailability.
- (f) Discuss the pH partition theory in detail with its limitations.
- (g) Describe the urinary excretion method for the estimation of elimination rate constant.